

MAD-MulW: A Multi-Window Anomaly Detection Framework for BGP Security Events

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Abstract—In recent years, various international security events have occurred frequently and interacted between real society and cyberspace. Traditional traffic monitoring mainly focuses on the local anomalous status of events due to a large amount of data. BGP-based event monitoring makes it possible to perform differential analysis of international events. For many existing traffic anomaly detection methods, we have observed that the window-based noise reduction strategy effectively improves the success rate of time series anomaly detection. Motivated by this, we propose an unsupervised anomaly detection model, MAD-MulW, which introduces a multi-window serial framework. The W-GAT module adaptively updates sample weights within the window to reduce noise, while the W-LAE module captures temporal trends through predictive reconstruction, enhancing inter-class separation. Our model has been experimentally validated on multiple BGP anomalous events with an average F1 score of over 90%, which demonstrates the significant improvement effect of the stage windows and adaptive strategy on the efficiency and stability of the timing model. The source code is available at://github.com/2024ChenYP/MAD-MulW.

Index Terms—Anomaly Detection, Time Series, Unsupervised Model, Multi-Window

I. INTRODUCTION

TODAY, the rapid development of the Internet has provided quality services in business, education, and entertainment. With society's increasing dependence on the Internet, its reliability and security have become critical concerns. Researchers have long focused on threats such as distributed denial-of-service attacks, network worms, and IP hijacking. Meanwhile, international conflicts (e.g., the Russia-Ukraine war) and natural disasters (e.g., earthquakes, hurricanes) can also trigger cascading network failures, causing incalculable losses. For instance, Hurricane Fiona struck Puerto Rico in 2022, damaging optical fiber and wireless networks and disrupting communication between Autonomous Systems (ASes). Although emergency services such as CISA and FEMA deployed mobile facilities, the disruption still had a significant impact. These examples highlight the necessity of accurate anomaly detection and warning to ensure network security.

In large-scale anomalous events, interface-level traffic monitoring is limited to a narrow view. As the most widely used inter-domain routing protocol, the Border Gateway Protocol (BGP) makes it possible to detect anomalies by managing

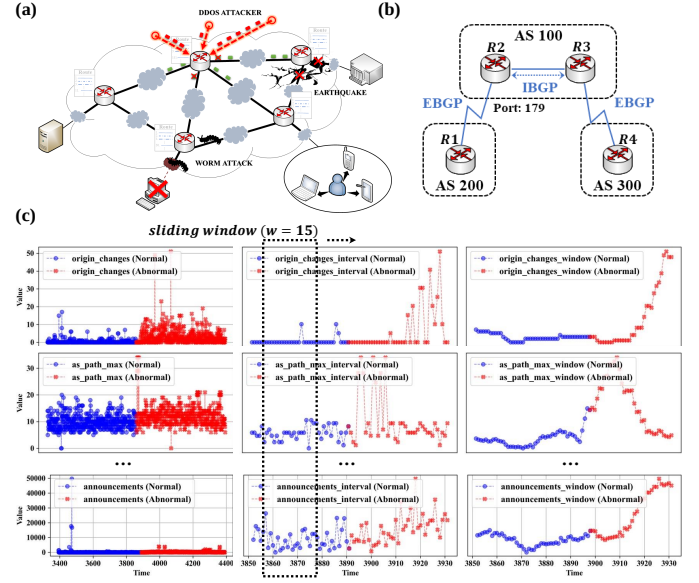


Fig. 1. (a) Illustration of different BGP anomaly events in the router network. (b) IBGP (internal) and EBGP (external) protocols both operate on port 179 for data exchange. (c) Time-series visualization of features under different processing methods: original, interval, and sliding window.

network reachability information. BGP anomalies are usually reflected in a surge of update messages when communication links are disrupted (Figure 1 (a)). By parsing and analyzing update packets from port 179 (Figure 1 (b)), one can derive time series suitable for anomaly detection. In this work, the investigated problem is defined as detecting abnormal timestamps from BGP update-derived multivariate time series under an unsupervised setting. The key challenge is that BGP security events disturb routing behaviors in different forms and intensities, while reliable anomaly labels are usually unavailable before detection. Therefore, the model is expected to learn normal BGP communication patterns and identify abnormal routing fluctuations through anomaly scores.

In terms of data, Semenoglou et al. [1] summarized the existing data enhancement methods and showed that the sliding window strategy is useful for most data analysis. Dimoudis et al. [2] proposed an anomaly detection algorithm for rolling median with an adaptive sliding window to detect anomalous data points in time series. These data enhancement strategies can smooth out the data noise and facilitate the task of classification, prediction, and detection. In terms of models, the field of time series provides many proven theories and techniques. The existing unsupervised methods are mostly based

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on three categories: Long Short Term Memory (LSTM)-used temporal models include LSTM-AD [3] and LSTM-GSVM [4], which specifically focus on the temporal attributes present in the samples. Additionally, sequence models structured with Autoencoders (AE) like TCN-AE [5] and LSTM-AE [6] create sequence mappings by encoding and decoding. Moreover, Generative Adversarial Network (GAN) models such as MTS-DCGAN [7] and MAD-GAN [8] play a role in generalizing the distribution of data. These methods have demonstrated excellent performance in a wide range of fields, including the Internet, industry, and medicine.

Sliding windows are widely used for data enhancement and model design (Figure 1 (c)), aiming to weaken noise through averaging or feature expansion. However, two major limitations remain: (1) no fixed optimal window size exists for different data distributions, and (2) window expansion increases computational cost. Moreover, when combined with LSTM-based models, it is difficult to provide interpretable and precise utilization of temporal properties. Therefore, it is worthwhile to explore generalizable and stable multi-stage window methods that can adapt to different anomaly scenarios.

Based on the above findings and ideas, we propose a stage-two window for feature remodeling and sequence prediction, respectively. Using anomalous event traffic data as input, we group them through a stage-one window and reshape the features using the Graph Attention Networks (GAT) module to reflect the state after noise reduction with the tail samples of the window; we use the LSTM-AutoEncoder (LAE) module to internally predict and externally reconstruct the data through window grouping in the second stage to achieve stable inter-class differences. The overall purpose is to generalize the strong similarity of feature fluctuation trends of normal samples. We take this similarity as a major basis for unsupervised anomaly detection, and the difference in anomalous samples will be expressed as high scores. We validate our model on several Internet traffic anomalies, and its performance is outstanding compared to many superior methods. In summary, the contributions of our study are mainly in the following:

- **Domain-focused perspective:** Unlike generic time-series anomaly detection, our work specifically addresses anomalous events in BGP communications, covering routing dynamics, prefix updates, and inter-domain fluctuations in real-world routing data. We formulate this problem in an unsupervised setting, where the absence of ground truth increases the challenge of learning abnormal patterns from unseen events.
- **Efficient multi-window design:** We propose Multivariate time-series Anomaly Detection with Multi-Window (MAD-MulW), which integrates W-GAT and W-LAE modules. W-GAT achieves adaptive noise reduction within windows, while W-LAE enhances detection by predictive reconstruction.
- **Stable and robust performance:** The proposed method achieves consistently superior detection performance across various anomalous events, improving the average F1 score by at least 20% compared to baseline models. Moreover, adaptive windowing contributes up to an 8% improvement in F1, while maintaining robustness under

parameter changes and limited training samples.

Subsequently, Section II focuses on the related work to anomaly detection. Section III presents a detailed explanation of our model. Section IV contains an introduction to the datasets, comparative models, and evaluation metrics. Section V evaluates the model and analyzes the experimental results. Finally, a summary of our work is presented in Section VI.

II. RELATED WORK

In this section, we describe the BGP anomaly detection work and extend it to the development of anomaly detection techniques for time series in recent years.

A. BGP Anomaly Detection Techniques

BGP is the Internet's default inter-domain routing protocol used to manage the connectivity between ASes. Over the past 20 years, many BGP events have been captured that threaten the stability of the Internet.

Supervised learning-based BGP anomaly detection has been extensively studied, and some of these methods affect anomaly detection performance mainly by improving the quantity or quality of features. Urbina Cazenave et al. [9] showed that Support Vector Machines (SVM), Decision Trees, and Naive Bayesian methods are evaluated for BGP event classification. Arai et al. [10] and Xu et al. [11] extracted features by an importance assessment algorithm to enhance the model efficiency. Innovatively, Sanchez et al. [12] introduced graph features to detect BGP anomalies, which are more robust than traditional features. Another class of methods focused on sample temporal relationships. The stacked-LSTM [13] and the MSLSTM [14] both achieved significant improvement in anomaly detection performance by mining temporal information. Peng et al. [15] went further by introducing a multi-dimensional attention mechanism to improve performance. The supervised approach requires the labeling of the dataset, so its applicability is limited to specific instances.

Unsupervised learning-based work is more sparse. Earlier statistical-based methods used probabilistic models to detect changes in the data, such as Principal Component Analysis (PCA) [16], Generalized Likelihood Ratio Test [17] and t-test [18]. In recent years, Andrian et al. [19] implemented DenStream, an anomaly detection engine based on clustering techniques, and applied it to a large testbed consisting of dozens of routers. Tal et al. [20] proposed the AP2Vec method, which embeds both ASes and IP address prefixes into feature vectors for BGP hijacking detection. BGP anomaly detection still has a lot of room for research in unsupervised anomaly detection techniques.

B. Time Series Unsupervised Anomaly Detection

Network traffic anomalous events can be viewed as anomaly detection problems under multivariate time series, and the design of unsupervised methods for this problem is highly challenging and has been widely studied in many fields.

Classical unsupervised methods are widely used for time series data anomaly detection, distance-based (K-Nearest

Neighbor, KNN [21]) methods modeled the local behavior around each data point, density-based (Local Outlier Factor, LOF [22]) methods discovered outliers by judging the density around the object domain, clustering-based (Gaussian Mixture Model, GMM [23]) methods grouped similar data points, and projection-based (PCA [24]) and classification-based (One-class SVM, OCSVM [25]) methods performed spatial separation of the original data by a linear transformation to achieve anomaly detection.

Deep learning-based unsupervised methods have numerous perspectives to model inter-metric dependencies. Convolutional networks have been applied to sequences for decades, and Bai et al. [26] proposed a Temporal Convolutional Network (TCN) that focuses more on the importance of temporal properties. On modeling inter-metric dependencies, Thill et al. [3] and Ergen et al. [4] used LSTM for anomaly detection of ECG and HTTP data after preprocessing the data with sliding windows for dimensionality enhancement, and designed window-based error correction methods to optimize the results. AE detects anomalies by deviations between encoded and decoded data of time series. Mo et al. [5] reconstruct time series using TCN-AE and Hou et al. [6] use LSTM-AE for higher-order feature learning, both of which are groundbreaking in anomaly detection. A relatively new area is the application of GAN [27]. Similarly, Liang et al. [7] proposed a new multi-timescale deep convolutional generative adversarial network (MTS-DCGAN) framework for industrial time series anomaly detection. Based on the several approaches mentioned above, most of the existing studies focused on enhancement and fusion. DAGMM [28] jointly optimized the parameters of the deep autoencoder and the mixture model simultaneously in an end-to-end fashion. MTAD-GAT [29] combined windows with GAT to learn complex dependencies of multivariate time series in time and feature dimensions and jointly predicted and reconstructed models to achieve detection functions. MAD-GAN [8] took window subsequences with different resolutions as input and captured the temporal correlation of time series distribution using LSTM-RNN. OmniAnomaly [30] identified anomalies using key techniques such as random variable concatenation, planar normalized flow, and robust representation after segmenting the sequence with a window. More recently, D3R [31] addressed the challenge of unstable data by dynamically decomposing long-period time series and employing diffusion-based reconstruction to control the information bottleneck externally, which effectively reduces false alarms caused by distribution drift.

In addition, several recent studies have further explored advanced representation learning, uncertainty-aware reconstruction, and diffusion-based generative modeling for time-series anomaly detection. Feng et al. [32] proposed Sensitive-HUE for multivariate time-series anomaly detection, which enhances the sensitivity to normal patterns by combining reconstruction with heteroscedastic uncertainty estimation. Su et al. [33] proposed an enhanced recurrent convolutional encoding method with attention-based representation learning for chaotic time series anomaly detection, where empirical mode decomposition, recurrent convolutional encoding, and attention mechanisms are integrated to suppress noise and capture

nonlinear temporal dependencies. Zuo et al. [34] explored unsupervised diffusion-based anomaly detection for time series, showing the potential of diffusion models in learning normal temporal patterns and identifying abnormal deviations. More recently, Pan et al. [35] proposed CT-DDPM, a copula and Transformer-based denoising diffusion probabilistic model for multivariate time-series anomaly detection, which uses Gaussian Copula to model multivariate joint distributions and Transformer-based denoising diffusion to capture temporal dependencies and distributional deviations. Liu et al. [36] further summarized recent advances in diffusion models for anomaly detection, indicating that diffusion-based anomaly detection has become an emerging research direction. In Table I, we compare the different methods from four perspectives: feature augmentation, time dependence, feature correlation, and interpretability. The more targeted mining analysis of these techniques improves the performance of each domain further while also observing the high frequency of windows in both data analysis and model construction. The performance of such methods depends heavily on the length of the time window and, therefore, requires targeted window lengths to provide satisfactory performance. Some of the methods consider data with different window sizes simultaneously in a multi-scale manner, which brings the problem of data dimension multiplication along with stable performance. The relationship between performance and consumption needs to be considered.

TABLE I
METHOD COMPARE

Model	Data Augmentation	Time Dependence	Feature Correlation	Interpre.
TCN-AE	Data scaling	✓		
LSTM-AE	/	✓		
MTS-DCGAN	Multi-time scale sliding windows		✓	✓
MTAD-GAT	Sliding windows	✓	✓	✓
MAD-GAN	Sliding windows	✓		
OmniAnomaly	Sliding windows	✓		✓
D3R	Disturbance	✓	✓	✓
Ours	Adaptive sliding windows	✓	✓	✓

Undoubtedly, the main trends of existing methods focus on temporal properties guided by LSTM and are concerned with reconstructive adversarial methods, mainly AE and GAN. Meanwhile, recent cutting-edge methods further extend this trend toward attention-guided representation learning, uncertainty-aware reconstruction, multivariate distribution modeling, and diffusion-based generation. These methods improve anomaly detection from different perspectives, but they are mainly designed for general time-series or chaotic time-series scenarios. At the same time, facing the problems of irregular size and increased resource consumption brought by the window strategies found above, we expect to design adaptive window optimization models to achieve efficient and generalized stable anomaly detection performance.

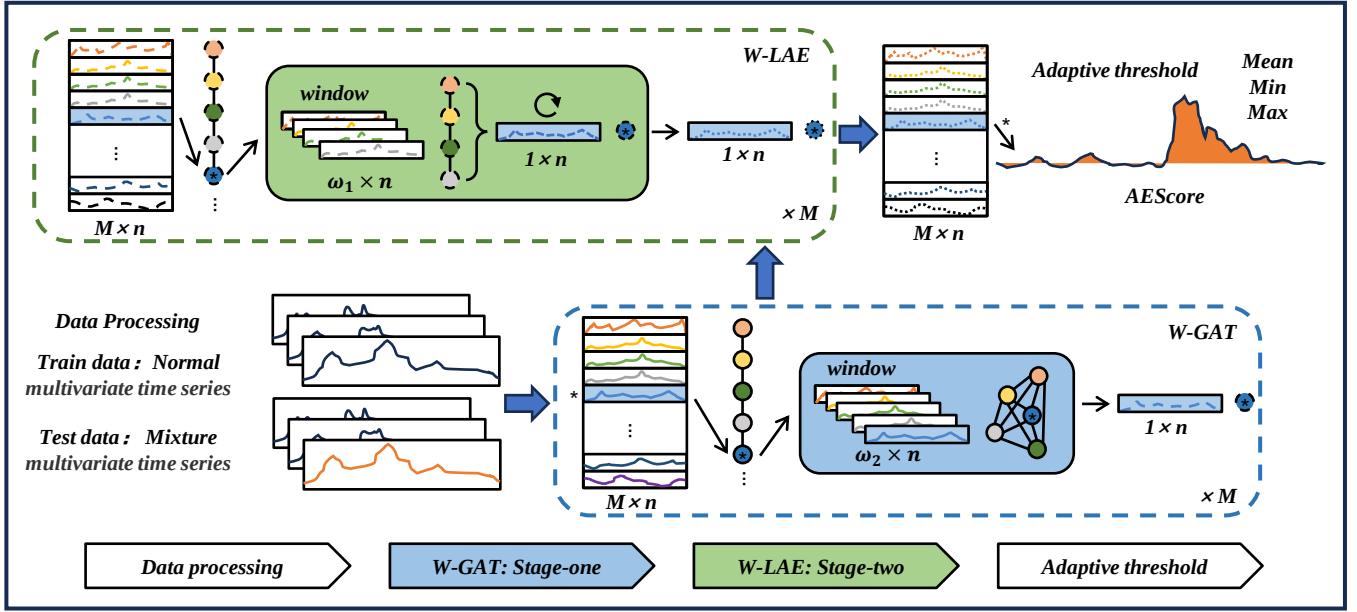


Fig. 2. Model Framework, MAD-MulW. The model has four main components: data processing, stage-one window, stage-two window, and adaptive threshold strategy. Among them, we indicate in detail the learning of sample relationships and retention of the last sample by GAT (blue); and the reconstruction and prediction of samples by LSTM-AE (green). Finally, the sample scores are obtained without changing the data dimensionality to achieve anomaly detection.

III. METHODOLOGY

We propose the unsupervised anomaly detection model MAD-MulW for the current problems of low feature utilization, high model fluctuation, and weak event generalization in the field of BGP anomaly detection. The overall structure is shown in Figure 2.

A. Problem Definition

Our study concentrates on diverse large-scale network security events, specifically traffic monitoring and characterization within a specified range. In practical inter-domain routing systems, abnormal events such as worm propagation, power failures, prefix leaks, and other network disruptions may cause sudden and irregular changes in BGP update messages. These changes are reflected in routing-related measurements, including announcement and withdrawal variations, prefix dynamics, AS-path changes, and other communication-flow features. Therefore, the actual problem addressed in this paper is to identify abnormal timestamps whose routing behaviors significantly deviate from normal BGP communication patterns. This paper focuses on features of the communication flow change information, which can be regarded as a multivariate time series of unsupervised anomaly detection. The input data is defined as $\mathcal{T} = \{x_1, x_2, \dots, x_M\}$, $x \in \mathbb{R}^n$, M represents the number of time series samples, and n represents the feature dimension corresponding to each time series sample. Therefore the t th sample can be represented as $x_t = \{x_t^1, x_t^2, \dots, x_t^n\}$. It should be noted that the univariate setting $n = 1$ is a special case of multivariate time series [37].

We define the binary variable $y \in \{0, 1\}$, where y_t is labeled as 0 if no unusual event occurs at timestamp t and 1 otherwise. However, in real BGP monitoring scenarios, accurate anomaly labels are usually unavailable before detection, and anomalous

events may appear with different durations, intensities, and routing impacts. To address the problem of unsupervised anomaly detection, we study the data features associated with anomaly detection. The training input $\mathcal{T}$ should consist solely of normal samples, denoted as $y_t = 0$ for $1 \leq t \leq M$. After learning the normal temporal fluctuation patterns of BGP communication features, a score of a trained model measures the difference between the invisible sample $\hat{x}_t, t > M$ (All subsequent vectors with the symbol $\hat{\cdot}$ represent invisible samples.) and $\mathcal{T}$. The normal or abnormal labels of the invisible sample are obtained by adaptive threshold strategy. In this way, the proposed formulation is consistent with the actual BGP anomaly detection scenario: the model learns stable normal routing behaviors from historical BGP update features and detects abnormal timestamps according to their deviations from learned normal patterns, without relying on labeled anomalous samples during training.

B. Data Processing

Standard data processing methods include data cleaning, data normalization, and feature engineering, among others. We process time series datasets in a targeted way to enhance subsequent model learning.

First, we check the data for missing values. If any are found, we fill the vacant part with feature values from the previous time step. Subsequently, we manually clean the columns containing unlearnable features. If the time series' order already captures moment-specific information, the corresponding feature becomes redundant and can be cleaned. Then, since the dataset comes with labels and our model is an unsupervised model, we separate the features and labels. Finally, the dataset is divided into a training set with all normal samples and a testing set with both normal and abnormal samples in a certain proportion.

C. Multi-window Structure

Time series are characterized by their inherent temporal continuity. On the one hand, this property can be exploited to re-represent time series and construct datasets with enhanced temporal properties. Thus, we propose and design a stage-one window module W-GAT, which employs self-attentive generation to optimize the dataset. On the other hand, we incorporate a stage-two window module W-LAE that replaces individual time points with groups. This is achieved by utilizing the LSTM model in conjunction with the AE model, enabling prediction and anomaly detection in time series data.

1) **Stage-one window W-GAT:** Time series representation is a process of reshaping the data, which facilitates the completion of subsequent model tasks and more significantly improves model efficiency. Ding et al. [38] used GAT to explicitly model the different relationships in multimodal time series to obtain a better representation of the input data. We designed a stage-one window module W-GAT for reshaping the representation of time series data, which consists of two parts: the first part is a stage-one window design, and the second part is a self-attentive model, as shown in Figure 3.

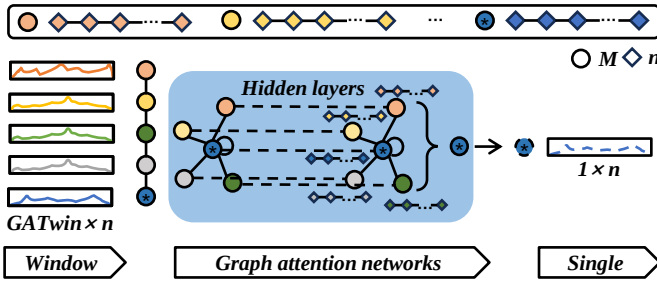


Fig. 3. stage-one window W-GAT.

Stage-one window group: The first part applies to all datasets. We design a uniform window size parameter w_1 and wish to represent a timestamp x_t with the past timestamps $[x_{t-w_1+1}, \dots, x_{t-1}, x_t]$. To avoid ambiguity between the original sample and the remodeled representation, we denote the W-GAT output at timestamp t as $\tilde{x}_t$. A fully connected network is constructed using graph nodes to represent each timestamp in the window and weights α to indicate the degree of association. The final theoretical representation is

$$\tilde{x}_t = \sum_{i=0}^{w_1-1} \alpha_{t,i} x_{t-i}, \quad \sum_{i=0}^{w_1-1} \alpha_{t,i} = 1 \quad (1)$$

where $\alpha_{t,i}$ denotes the attention weight assigned to the i -th historical sample within the stage-one window at timestamp t .

Graph attention networks: It is difficult to capture the optimal situation of different datasets by manually adjusting the parameter α . So we utilize the self-attention network to correlate the time series and obtain the timestamp attention weights automatically.

The input of stacking graph attention layers is a set of timestamp feature vectors, $[x_{t-w_1+1}, \dots, x_{t-1}, x_t]$. For a node i in the window graph, the updated representation is defined as:

$$h_i = \sigma \left(\sum_{j \in N_i} \alpha_{ij} W x_j \right) \quad (2)$$

where W is the weight matrix, $\sigma(\cdot)$ is the nonlinear activation function, N_i represents the set of all neighbors of timestamp i , and α_{ij} is attention coefficient in GAT which is calculated as:

$$\alpha_{ij} = \frac{\exp(LR(a^T [W x_i \| W x_j]))}{\sum_{k \in N_i} \exp(LR(a^T [W x_i \| W x_k]))} \quad (3)$$

The updated node representation h_i is a new feature representation for timestamp i , which reinforces the influence of historical feature information on the current timestamp. For each stage-one window ending at timestamp t , the final W-GAT-remodeled feature $\tilde{x}_t$ is obtained from the output representation of the current timestamp node in the window.

The stage-one window W-GAT plays a role in reshaping the features in the overall model and is the key point of this paper. Unlike previous work that utilized GAT to capture strong and weak correlations at the time level or feature level, this section uses samples as the updating unit and focuses only on the degree of influence of all historical samples on the most recent sample. By using GAT to adaptively update the weights, the stability of the input information is improved by smoothing the current feature fluctuations with historical information. At the same time, it ensures that the dimension of the output data is equal to the dimension of the original data, avoiding the increase in space cost brought by GAT.

2) **Stage-two window W-LAE:** The focus of the stage-two window is on data reconstruction based on time-series prediction. Distinguishing from traditional autoencoders, our approach specifies a window size of historical samples, predicts the sample information at the next moment, and reconstructs the individual timestamps using the property of multi-featured variables, i.e., the size of the encoder's inputs is not equal to the size of the outputs. This design captures the temporal relationship between samples and also increases the variability between different classes of samples to achieve improved detection performance. The structure of stage-two window W-LAE is shown in Fig 4.

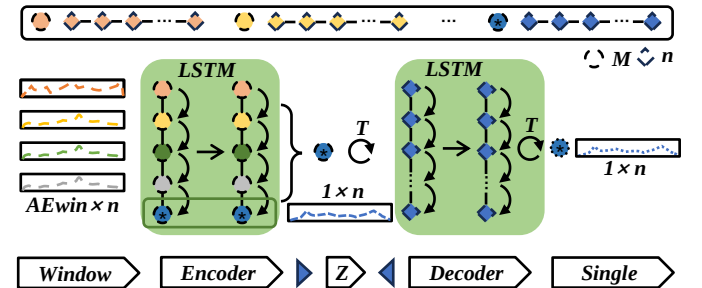


Fig. 4. stage-two window W-LAE.

Stage-two window group: The first part of the window group is designed to allow multiple previous timestamps to replace the current timestamp. In this case, we set the stage-two window parameter w_2 . Based on the remodeled sequence

$\tilde{T} = \{\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_M\}$, the input window for predicting timestamp t is defined as:

$$\mathbf{X}_t^{(w_2)} = [\tilde{x}_{t-w_2}, \tilde{x}_{t-w_2+1}, \dots, \tilde{x}_{t-1}] \in \mathbb{R}^{w_2 \times n} \quad (4)$$

Note that here the window group represents a certain timestamp but does not contain the time series at that time so that the purpose of predicting the unknown time series from known previous data can be achieved.

The role of the encoder is to map the historical window input $\mathbf{X}_t^{(w_2)}$ into the latent representation $\mathbf{z}_t$ through a nonlinear transformation.

$$\mathbf{z}_t = f_{\text{enc}}(\mathbf{X}_t^{(w_2)}; \boldsymbol{\theta}_{\text{enc}}) \quad (5)$$

where $\mathbf{X}_t^{(w_2)} = [\tilde{x}_{t-w_2}, \dots, \tilde{x}_{t-2}, \tilde{x}_{t-1}]$ denotes the stage-two historical window input, and $\boldsymbol{\theta}_{\text{enc}}$ denotes the parameters of the LSTM encoder.

The role of the decoder is to predict the feature representation at timestamp t from the latent variable $\mathbf{z}_t$. The mapping process can be expressed as follows:

$$\hat{x}_t = f_{\text{dec}}(\mathbf{z}_t; \boldsymbol{\theta}_{\text{dec}}) \quad (6)$$

where $\hat{x}_t \in \mathbb{R}^n$ is the predicted representation of the current timestamp, and $\boldsymbol{\theta}_{\text{dec}}$ denotes the parameters of the decoder. For normal samples, the prediction output $\hat{x}_t$ is expected to be close to the corresponding W-GAT-remodeled feature $\tilde{x}_t$, i.e., $\hat{x}_t \approx \tilde{x}_t$.

The main feature of the W-LAE module in this part is that the neural network used for the mapping process is fixed as LSTM. The principle of training W-LAE is to minimize reconstruction errors. On the one hand, it retains the strong generalization of the reconstruction-based approach under unsupervised tasks, and on the other hand, it also preserves the information on the relationship between features in the process of encoding and decoding. Meanwhile, the hidden variable $\mathbf{z}_t$ can be regarded as an important feature extracted from the input data that is sufficient for subsequent analysis. All these factors guarantee the excellent performance of W-LAE in the unsupervised anomaly detection task, which will be verified in the subsequent experiments.

D. Our Model

Our unsupervised anomaly detection model is mainly divided into a training part and a testing part, and these parts of the model are made to unite into a whole by designing suitable loss functions and score functions by ourselves.

The original dataset has been cleaned up after data processing. We completed the data reshaping through a stage-one window W-GAT, which step changed the data feature values in the time dimension to reduce noise in continuous samples. After that, the prediction task is implemented by the W-LAE module with data grouping and data reconstruction. For each timestamp t , the W-LAE module takes $\mathbf{X}_t^{(w_2)}$ as input and outputs the predicted feature vector $\hat{x}_t$. The deviation between $\hat{x}_t$ and the corresponding W-GAT-remodeled feature $\tilde{x}_t$ is used to measure the abnormality of timestamp t . Specifically, the timestamp-level anomaly score is defined as follows:

$$s_t = \frac{1}{n} \|\hat{x}_t - \tilde{x}_t\|_1 = \frac{1}{n} \sum_{d=1}^n |\hat{x}_t^d - \tilde{x}_t^d| \quad (7)$$

where n is the feature dimension and d indexes each feature. A larger s_t indicates a larger deviation from the learned normal temporal pattern.

The reconstruction error is returned as a loss function during the training process, to make the prediction result similar to the actual timestamp. Given N valid training timestamps after window construction, the training objective is formulated as follows:

$$\mathcal{L} = \frac{1}{N} \sum_{t=1}^N s_t = \frac{1}{N} \sum_{t=1}^N \frac{1}{n} \|\hat{x}_t - \tilde{x}_t\|_1 \quad (8)$$

During testing, the timestamp-level anomaly score s_t is used for adaptive threshold-based anomaly detection. The Mean and Standard Deviation approach is a commonly used statistical method to set adaptive thresholds based on the distribution of the model's output loss values.

Calculation of Mean and Standard Deviation: Let $\mathcal{S}_{\text{train}} = \{s_1, s_2, \dots, s_N\}$ denote the set of anomaly scores computed from the normal training samples, where N is the number of valid training timestamps after window construction. The mean (μ) and standard deviation (σ) of the anomaly scores are calculated as follows:

$$\mu = \frac{1}{N} \sum_{t=1}^N s_t \quad (9)$$

$$\sigma = \sqrt{\frac{1}{N} \sum_{t=1}^N (s_t - \mu)^2} \quad (10)$$

Threshold Determination: The anomaly detection threshold is defined by setting a threshold level that incorporates a multiple of the standard deviation added to the mean. This is expressed as:

$$\delta = \mu + k \times \sigma \quad (11)$$

where k is a scaling factor that determines the sensitivity of the threshold to variations in the loss distribution. The value of k should be selected based on the specific application and acceptable risk levels of false positives and negatives. In the BGP anomaly detection task, $k = 25$.

For each new data point, the corresponding loss value is computed and compared against the established threshold. For an unseen sample, if its anomaly score satisfies $s_t > \delta$, it is identified as abnormal; otherwise, it is regarded as normal:

$$\hat{y}_t = \begin{cases} 1, & s_t > \delta, \\ 0, & s_t \leq \delta. \end{cases} \quad (12)$$

Following multiple training rounds and optimization based on the complete process described above, the model performance can be validated on a test set.

IV. EXPERIMENTAL SETUP

A. Datasets

The datasets used in this thesis are derived from several classes of classical BGP anomalous events [39]. We summarize the information of each dataset in Table II, and its detailed description is shown below.

TABLE II
BGP ANOMALY EVENT DATASETS

Dataset	Total	Anomaly (Rate)	Features	Time or Field
Code Red II	7136	472 (6.61%)	48	2001.07.17-2001.07.21
Nimda	10336	353 (3.42%)	48	2001.09.15-2001.09.23
Slammer	7200	1130 (15.69%)	48	2003.01.23-2003.01.27
Moscow	7200	171 (2.38%)	48	2005.05.23-2005.05.27
Malaysian	7200	185 (2.57%)	48	2015.06.10-2015.06.14

Code Red II: On July 19, 2001, the Code Red II worm started its spreading across the global network. Since the beginning of the anomaly, an exponentially growing eight-fold increase in the BGP advertisement rate was observed over a period of about eight hours.

Nimda: On September 18, 2001, as the Nimda worm started its propagation, Over a period of roughly two hours, the rrc00 collector perceived BGP advertisement rates exponentially ramped up by a factor of 25. The advertisement rate then decayed gradually over several days.

Slammer: The Slammer worm was released on Jan 25, 2003, 5:31 UTC, infected at least 75,000 hosts in just over 30 minutes and it was reported that a number of critical AS-AS peering links were operating above critical load thresholds during the attack period.

Moscow Blackout: The Moscow Power Blackout occurred on May 25, 2005, and lasted several hours. The effect was apparent at the RIS remote route collector in Vienna (rrc05) through a surge in announcement messages arriving from peer AS 12793.

Malaysian Telecom Leak: Malaysian Telecom (AS 4788) leaked one-third of the IP prefixes in its global routing table to backbone provider Level 3 (AS 3549) from June 12, 2015. It left the company inundated with data, resulting in severe packet loss and performance degradation.

By using the event time period and the ASes involved, complete BGP update data can be obtained from the Route Views [40] and RIPE NCC [41] websites. Several statistical features can be constructed by parsing the packets and extracting specific information. Our dataset contains 48 features, which are divided into two categories: *Volume features* and *AS-path features*, and described in Appendix Table VI.

B. Comparative Methods

To ensure comprehensive coverage of the comparison methods, we carefully selected 11 techniques that encompass a wide range of unsupervised approaches, including both

classical and the latest methods. Among them, the **classical methods** include KNN, Cluster-based Local Outlier Factor (CBLOF), Histogram-based Outlier Score (HBOS), Isolation Forest (iForest), OCSVM, and Principal Component Analysis (PCA). The **latest methods** include Deep Autoencoding Gaussian Mixture Model (DAGMM) [28], Multivariate Time-series Anomaly Detection via Graph Attention Network (MTAD-GAT) [29], Multivariate Anomaly Detection with Generative Adversarial Network (MAD-GAN) [8], OmniAnomaly [30], and D3R [31], and these methods have been widely approved and used in time series anomaly detection work in recent years.

C. Evaluation Metrics

We view the BGP anomaly detection as a classification problem, thus choosing *Accuracy*, *Precision*, *Recall*, and *F1* as the evaluation metrics. The dataset of real anomalous events exhibits a significant class imbalance. Among them, *F1* is a weighted average of model precision and recall that can be used to measure the precision of unbalanced data. To facilitate comparison, we assign the abnormal samples as the positive class and the normal samples as the negative class. This allows classification into True Positive (TP), False Positive (FP), True Negative (TN), or False Negative (FN).

According to the above four situations, we can calculate each performance metric as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (13)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (14)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (15)$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (16)$$

D. Model Parameters

The performance of a model is sensitive to variable parameters. We split the normal data into a training set and a testing set with a ratio of 7:3, the training set consists of normal data and the test set contains normal and abnormal data. The training parameters are chosen as follows: learning rate is $1e-2$, the epoch is 10, the W-GAT window is 15, and W-LAE is 11. Part of the parameter setting of the model is adjusted according to the aggregation of the concrete experiment. This selection can help the model achieve excellent and stable results on many datasets. Our model and all experiments are implemented on *Python*, relying on the *PyTorch* framework, the *sklearn* library, and other related libraries and functions.

V. EXPERIMENTS AND RESULTS

This section focuses on evaluating the model's performance in unsupervised anomaly detection and assessing the effectiveness of the stage-two window design for each structure component. We then discuss the rationality of each of the three hyperparameter designs separately.

TABLE III
EXPERIMENT RESULTS

Dataset	Model	Classical Model						Latest Model					Ours
		KNN	CBLOF	HBOS	iForest	OCSVM	PCA	MTAD-GAT	MAD-GAN	DAGMM	OmniAnomaly	D3R	
Code Red II	Acc.	85.29	85.03	85.31	87.22	84.87	85.85	88.96	76.62	94.96	90.65	81.53	97.82
	Pre.	51.62	50.92	51.69	56.99	50.45	53.17	66.11	64.96	65.57	65.81	79.98	96.54
	Rec.	52.36	51.34	52.47	60.18	50.65	54.63	81.33	64.04	50.53	72.79	52.87	96.13
	F1	51.72	50.89	51.81	57.07	50.34	53.56	70.23	64.46	57.14	68.43	50.54	96.33
Nimda	Acc.	65.78	64.48	66.75	70.69	64.43	67.37	69.75	72.72	67.41	68.09	39.54	86.76
	Pre.	58.72	55.12	61.41	72.37	54.96	63.13	66.28	65.48	69.41	66.12	62.96	86.21
	Rec.	53.49	52.05	54.56	58.95	51.99	55.25	66.06	70.66	8.35	65.34	52.28	85.04
	F1	50.31	48.43	51.72	57.45	48.34	52.61	66.17	66.28	14.90	65.60	32.64	85.55
Slammer	Acc.	83.06	82.14	84.36	87.47	82.03	83.78	83.63	77.32	83.19	80.20	62.11	98.19
	Pre.	65.59	63.04	69.21	77.85	62.73	67.59	69.54	70.20	38.24	69.43	74.47	98.12
	Rec.	60.60	58.87	63.07	68.95	58.66	61.97	70.79	76.58	11.50	79.97	51.08	97.93
	F1	62.17	60.12	65.08	72.03	59.87	63.78	70.07	71.65	17.69	71.72	40.51	98.02
Moscow	Acc.	91.99	91.90	91.99	92.12	91.74	91.99	98.24	73.38	97.92	98.06	95.34	99.36
	Pre.	60.79	60.56	60.79	61.18	60.10	60.79	80.92	66.16	53.85	81.13	81.00	99.67
	Rec.	91.90	91.00	91.90	93.40	89.21	91.90	80.92	71.55	100.00	71.13	97.21	91.03
	F1	65.49	65.13	65.49	66.08	64.41	65.49	80.92	67.06	70.00	75.11	86.91	94.90
Malaysian	Acc.	91.71	91.51	91.51	92.26	90.96	91.99	97.13	83.52	96.94	95.97	96.24	99.26
	Pre.	60.46	59.92	59.92	62.00	58.37	61.23	70.79	76.80	45.45	66.41	84.54	99.62
	Rec.	87.60	85.65	85.65	93.14	80.10	90.36	69.13	70.17	94.59	75.83	96.48	90.24
	F1	64.80	63.98	63.98	67.16	61.62	65.98	70.02	72.48	61.40	69.94	89.36	94.40
Average	Acc.	83.57	83.01	83.98	85.95	82.81	84.20	87.54	76.71	88.08	86.59	74.95	96.28
	Pre.	59.44	57.91	60.60	66.08	57.32	61.18	70.73	68.72	54.50	69.78	76.59	96.03
	Rec.	69.19	67.78	69.53	74.92	66.12	70.82	73.65	70.60	52.99	73.01	69.98	92.07
	F1	58.90	57.71	59.62	63.96	56.92	60.28	71.48	68.39	44.23	70.16	59.99	93.84

A. Basic Experiment

In this section, we compare the performance differences of numerous unsupervised anomaly detection methods on BGP anomaly traffic datasets, and the results are shown in Table III. Under each metric, the optimal results are bolded and underlined. Due to the small proportion of anomaly samples in datasets, detection becomes significantly more challenging. Consequently, the F1 metric provides a more comprehensive evaluation of the model's performance in both normal and anomalous sample identification. From the experimental results, we can clearly see that the latest methods generally outperform the classical detection models, mainly because the classical methods focus more on the partial attributes of events. Although more efficient, they are not as comprehensive as the latter in terms of detection performance after all. However, despite their general applicability, these latest models exhibit considerable performance fluctuations across different BGP anomaly events, indicating limited adaptability to event-specific characteristics.

We calculate the average of each model metric at the end of Table III, and the results clearly show that our model improves by over 20% on the most critical F1 metric compared to the best baseline and consistently achieves above 90% on accu-

racy, precision, and recall. To further present the results from a statistical perspective, we analyze the F1 scores across the five BGP anomaly datasets. Our method achieves an average F1 score of $93.84\% \pm 4.84\%$, while the strongest baseline in terms of average F1 score, MTAD-GAT, achieves $71.48\% \pm 5.55\%$. This indicates an absolute improvement of 22.36 percentage points and a relative improvement of 31.28%. Moreover, when comparing our method with the best-performing baseline on each individual dataset, the F1 improvements on Code Red II, Nimda, Slammer, Moscow, and Malaysian are 26.10, 19.27, 25.99, 7.99, and 5.04 percentage points, respectively. The improvement is positive on all five datasets, with an average gain of 16.88 ± 9.91 percentage points. These statistical results demonstrate that the proposed method achieves consistent and substantial improvement across different BGP anomaly events, rather than only a slight numerical increase. More importantly, our approach demonstrates strong stability across all tested events, benefiting from its capability to capture BGP-specific anomaly features.

B. Window validity

We have two window parameters, where the W-GAT window focuses on enhancing temporal correlations and feature

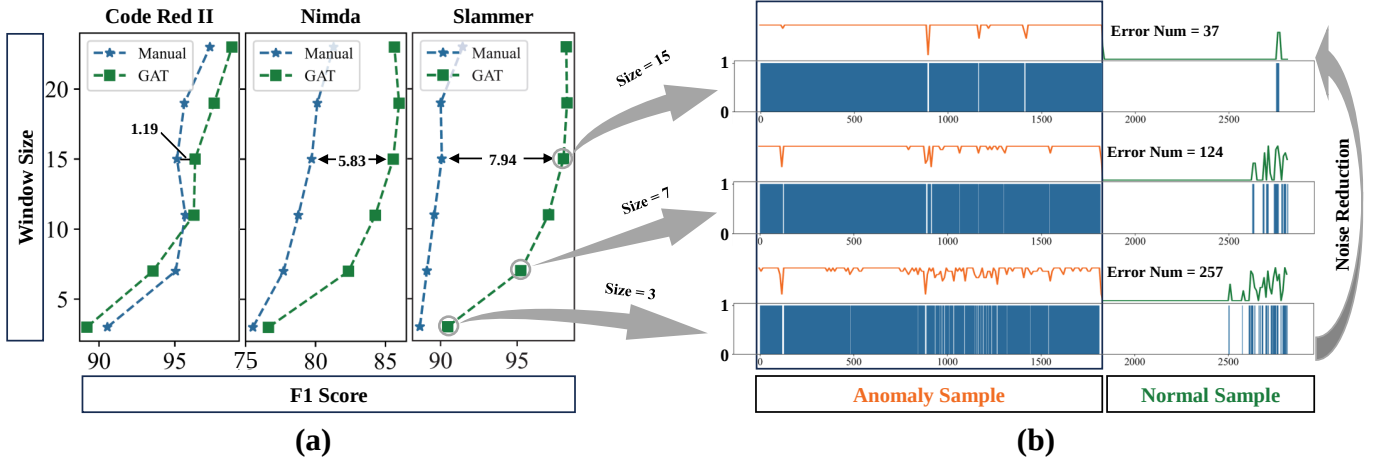


Fig. 5. W-GAT module validation. (a) Performance differences between the manual and adaptive windows; (b) The predicted labels of the test samples under different window sizes.

reconstruction, and the W-LAE window is responsible for prediction. To validate their effectiveness, we design experiments by selectively enabling or disabling these two components. Specifically, the W-LAE window is verified by comparing cases with and without W-LAE, while the W-GAT window is further examined under multiple conditions: (i) w/o W-GAT, (ii) w/ Manual W-GAT, and (iii) w/ Adaptive W-GAT. When the W-GAT window size is set to 15 and the W-LAE window size is set to 11, the experimental results on three datasets are shown in Table IV. The findings indicate that enabling only W-LAE (without W-GAT) achieves moderate performance improvement compared to using only W-GAT, confirming the necessity of both windows. Introducing a manual W-GAT window (with W-LAE) results in a 4%-11% increase in the F1 score, while the GAT-based adaptive window (with W-LAE) achieves the best performance, improving the F1 score by an additional 1%-6% and reaching 96.33%, 85.55%, and 98.02% for Code Red II, Nimda, and Slammer, respectively. These results clearly demonstrate that (i) both W-GAT and W-LAE windows contribute significantly to performance gains, and (ii) adaptive W-GAT design provides the most effective enhancement by dynamically optimizing temporal feature aggregation.

TABLE IV
WINDOW EXPERIMENT RESULTS

Window 1 (Size 15)	Window 2 (Size 11)	Dataset		
		Code Red II	Nimda	Slammer
w/ W-GAT	w/o W-LAE	88.64	68.43	81.02
w/o W-GAT	w W-LAE	86.31	73.46	86.32
w/ Manual W-GAT	w W-LAE	95.14	79.72	90.08
w/ Adaptive W-GAT	w W-LAE	96.33	85.55	98.02

However, is it a coincidence that the current window size makes excellent results? To answer this question, we obtain the experimental results in Figure 5(a). It is obvious from the three figures that the results of manual and adaptive are similar for small windows in the early stage, but as the window increases, the adaptive window shows a significant

performance improvement, even up to an 8% difference on the Slammer dataset. The optimal manual window value is not fixed for different datasets, and the impact of the window is inconsistent, showing a tendency to fluctuate upward with the window size. In contrast, the GAT adaptive window can determine the importance of the sample by the parameter value instead of fixing it to 0 or 1. This adaptive nature allows the F1 score to show a steady upward trend as the window size increases, ensuring that each dataset achieves near-optimal results in the GAT window.

A potential explanation is that the adaptive window of GAT simulates a manual window, using the connection weights between the time series within the window to simulate the window size, ultimately achieving a near-optimal result. In particular, the performance improvement of the window from presence to absence is the largest, and the explanation is also related to the presence of noise in the time series itself, as it is a non-ideal curve. The addition of the window effectively smoothes out the noise, eliminates outliers, and ensures overall smoothness. As shown in Figure 5(b), the fluctuations of the samples due to misclassification decrease significantly with the increase of the window size. Additionally, we artificially assign weights to the manual windows to simulate the adaptive windows of GAT, aiming to assess the impact of weights as opposed to fixed-value windows.

C. Parameter Validity

In the course of our experiments, three questions arose as follows:

- **Q1:** What is the effect of the number of training samples on the results?
- **Q2:** How does the threshold value affect the results?
- **Q3:** What are the stage-two window parameters set to?
- **Q4:** How does the model balance training, inference, size, and performance?

1) *The number of training samples:* The first question is due to the excellent performance and reasonable design of our model in the basic experiments described above, which makes

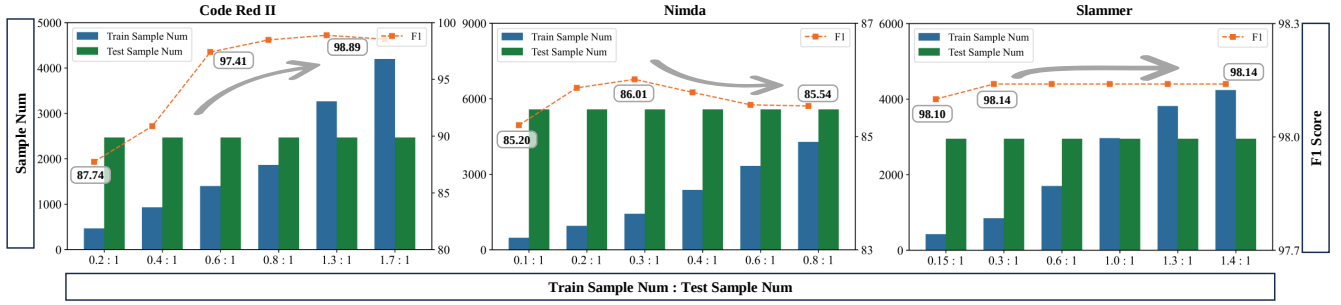


Fig. 6. The variation of the results with different sample ratios.

us look forward to further investigating whether the MAD-MulW model can achieve good detection performance with only a small number of training samples. So, there was a fixed number of test samples, and experiments were conducted for different numbers of training samples, and the experimental results are shown in Figure 6. The horizontal coordinates in the figure are the ratio of the number of train sets to the number of test sets, and the vertical coordinates correspond to the different metric values. The overall trend is more intuitive, and the detection performance improves as the number of train samples increases. It is worth noting that our model generally achieves optimal results using a train set with less than the number of test samples, which again demonstrates the stable and generalized detection performance of our model. The small amount of train data and high detection results indicate that the model requires low learning costs and fast learning. Notably, near-optimal performance is attained even with sparse samples, thereby reducing the restrictiveness of the anomaly detection task and enhancing its applicability in diverse situations.

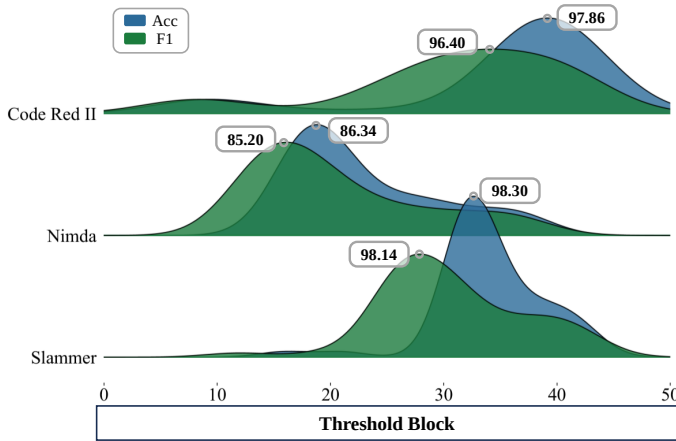


Fig. 7. The variation of the results with different thresholds.

2) *Adaptive threshold strategy*: The second question focuses on the study of the adaptive threshold strategy. Figure 7 shows the trend of the effect of different thresholds (horizontal coordinates) on the F1 score (vertical coordinates) of the three datasets. Both too-small and too-large thresholds lead to many misclassification cases, while the range of better thresholds is narrower, and the selection of thresholds varies on different datasets. Under our adaptive threshold strategy, the divided thresholds do not go through the lowest classification results

at the lowest and highest thresholds. Since the change of threshold value has little effect on the division at smoothing, the model can achieve good experimental results within a certain range of threshold values. The adaptive strategy clarifies the importance of threshold selection and also shows that achieving dynamic thresholds is a problem that needs to be studied in depth in subsequent work.

3) *Manual window design*: The third question is the setting of the stage-two window parameters. The trends of detection performance for different datasets with different window parameters are shown in Figure 8.

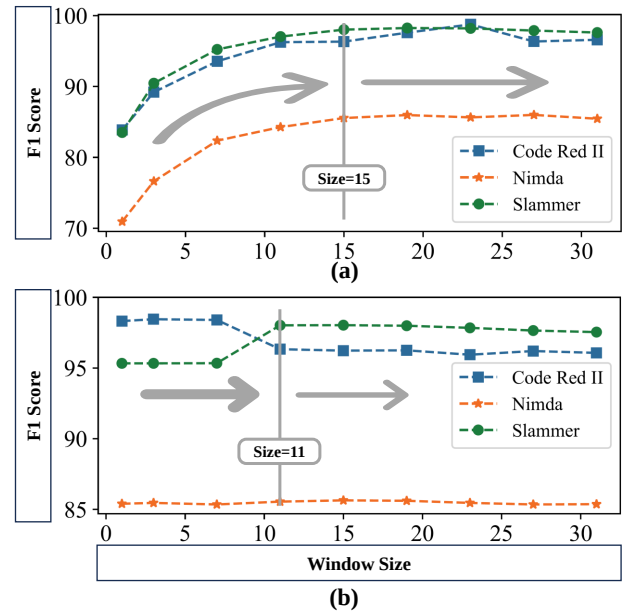


Fig. 8. The variation of the results with different window sizes.

Upon fixing the size of the AE window, we observed a consistent trend in the size of the GAT window across all datasets: a gradual increase followed by stabilization (Figure 8(a)). To ensure stable and outstanding results across multiple datasets while considering the minimum sample size, we fixed the window size at 15 for the W-GAT module.

The AE window serves to achieve prediction, so the size of the AE window directly affects the temporal memorability of the LSTM during computation. the LSTM needs to learn and infer information from features at known moments to unknown moments, while the long- and short-term dependence of the samples also determines the amount of information required.

Our experimental results found (Figure 8(b)) that when the AE window increases, the different datasets do not show a clear and uniform trend. However, when the window is at 11, a better state can be achieved by considering short-term temporal information.

4) *Model cost*: To verify the claim that our method achieves both computational efficiency and high detection performance, we conducted a comparative experiment to measure the time, space, and detection capability of our model against several baseline methods. Specifically, we evaluated four metrics: *Training Time*, *Inference Time*, *Model Size*, and *Avg. F1*. All experiments were carried out on the same hardware and BGP anomaly dataset to ensure fairness, allowing a direct comparison of computational efficiency and practical applicability.

TABLE V
COMPARISON OF COMPUTATIONAL COST, MODEL SIZE, AND DETECTION PERFORMANCE

Method	Training Time (s / epoch)	Inference Time (s)	Model Size (MB)	Avg. F1 (%)
MTAD-GAT	1.07	58.50	1.45	71.48
MAD-GAN	3.00	147.00	0.027	68.39
DAGMM	4.43	0.19	0.039	44.23
OmniAnomaly	167.68	32.04	8.45	70.16
D3R	4.02	31.78	12.70	59.99
Ours	18.86	16.03	0.80	93.84

Table V summarizes the results. Our method requires a moderate training time of 18.86s per epoch, which is longer than MTAD, MAD, DAGMM, and D3R but substantially shorter than Omni. More importantly, it achieves the fastest inference time among all methods (16.03s) while maintaining a compact model size of only 0.80MB. In terms of detection performance, our model reaches an average F1 score of 93.84%, which significantly surpasses all baselines, including MTAD (71.48%) and Omni (70.16%). These results indicate that our approach provides the best overall balance between inference efficiency, lightweight memory footprint, and high detection accuracy, demonstrating its strong practicality for real-world BGP anomaly detection scenarios.

So far, we have demonstrated the stability and generalization of our model design, as well as the stringency of the parameter design, through several experiments.

D. Result Interpretation

In parts A and B of this section, we validate the accuracy and validity of the model. In part C, we validate the sensitivity of the model involving the parameters, which is the most important validation and debugging technique of the model that maintains acceptable model behavior and predictions when the data is intentionally interfered with or simulated by other means [42]. In order to increase the credibility of the model's features and to gain a deeper understanding of the reasons behind the model's decisions, we have performed an interpretability analysis of the results in this subsection. Figure 9 jointly characterizes the data features with the model outputs,

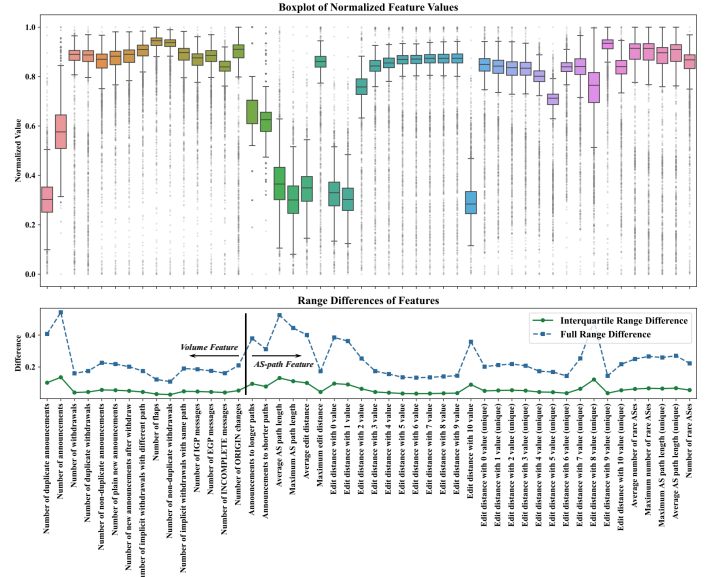


Fig. 9. The distribution of the normalized features and the differential of their distribution.

reflecting the sensitive relationship between the sample scores and the data. The horizontal coordinates of the box plot are the 48 different features, and the vertical coordinates are the distributions of the output vectors of the different features after being modeled (normalized due to the different scales of the different features). The wider distribution of a feature means that the feature is more sensitive to the variation of sample scores and is therefore regarded as a more important feature in the model output. Thus, we quantified the difference in the distribution of features using line plots, where the green solid line represents the interquartile range difference between the value represented by the upper quartile and the value represented by the lower quartile of the box plot, and the blue dashed line represents the full range difference between the upper and lower whisker limits of the box plot. It is obvious from the trend of the results that it is particularly important to focus on the volume features of the announcements and the path features of the ASes. This is in line with the mass broadcasting and alternative path adjustment when BGP anomalies occur. Meanwhile, the result also provides good guidance for the subsequent optimization of feature engineering.

VI. CONCLUSION

In this paper, we innovatively propose a stage-two window model for multivariate time series anomaly detection in order to achieve the detection and analysis of large-scale network anomalous events. The model employs a stage-one W-GAT window for feature reconstruction and a stage-two W-LAE window for time series prediction. Our proposed method outperforms baseline methods and advanced models on the BGP dataset. Furthermore, our model is characterized by low training costs and fast learning and achieves excellent detection results even with limited training samples. These characteristics also indicate the potential feasibility of MAD-MulW in resource-constrained router environments, since it

operates on extracted BGP update features, preserves the original feature dimension after W-GAT processing, and can be deployed as a lightweight monitoring component on the router control plane or an external route collector without directly affecting packet forwarding. Future work will further investigate model compression, quantization, and hardware-specific optimization for practical router-side deployment. Moreover, our method demonstrates applicability to a wide range of datasets thanks to its automatic reshaping capability and stable prediction performance.

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APPENDIX

FEATURE DESCRIPTION

A detailed description of the 48 features of the BGP anomaly event datasets.

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TABLE VI
FEATURE DESCRIPTION

No.	Feature name	Describe	Category
1	Number of announcements	General number of messages received that contain a request to include a BGP AS path in the Routing Information Base (RIB).	Volume
2	Number of withdrawals	General number of messages received that contain a request to remove a BGP AS path in the RIB.	Volume
3	Number of duplicate announcements	The mechanism keeps track of previously announced paths and marks consecutive announcements with the same attributes as duplicates.	Volume
4	Number of duplicate withdrawals	The mechanism keeps track of previously announced paths and marks consecutive announcements with the same attributes as duplicates.	Volume
5	Number of non-duplicate announcements	Announcements that contain information different from the stored in the RIB.	Volume
6	Number of non-duplicate withdrawals	Withdrawals that contain information different from the stored in the RIB.	Volume
7	Number of flaps	When a path is announced, succeeded by a withdrawal message that removes it from the RIB, then the same path is announced again with the same attributes.	Volume
8	Number of new announcements after withdrawal	When a path to a prefix is announced, then removed from the RIB, and then a path to the same prefix is announced with different attributes.	Volume
9	Number of plain new announcements	Announcements for a prefix that was never stored in the RIB.	Volume
10	Number of implicit withdrawals with same path	Consecutive announcements for the same prefix with the same path but with at least one different attribute (e.g. MED, LOCAL PREF, ORIGIN).	Volume
11	Number of implicit withdrawals with different paths	Consecutive announcements for the same prefix with a different path.	Volume
12	Number of IGP messages	The ORIGIN attribute indicates which protocol generated the BGP message, our mechanism keeps a counter for each ORIGIN type (e.g. IGP, EGP, INCOMPLETE).	Volume
13	Number of EGP messages		Volume
14	Number of INCOMPLETE messages		Volume
15	Number of ORIGIN changes	Counts when a re-announcement changes the ORIGIN attribute previously announced.	Volume
16	Announcements to longer paths	This feature tracks whether a new path is longer or shorter than the path previously stored.	AS-path
17	Announcements to shorter paths		AS-path
18	Average AS path length	Average AS path length considering the path announced in all BGP messages during a time window w .	AS-path
19	Maximum AS path length	Maximum AS path length found in received BGP messages during a time window w .	AS-path
20	Average AS path length (unique)	Some ASes artificially inflate the path length of an announced path by including redundant hops in a path (e.g., path = (A B B B C)). This strategy is widely used to decrease the chances of a route being chosen by a peer (e.g. backup path). This metric calculates the average AS path ignoring prepending and considering the paths announced in all BGP messages during a time window w .	AS-path
21	Maximum AS path length (unique)	Maximum AS path length, ignoring prepending, found in received BGP messages during a time window w .	AS-path
22	Average edit distance	When an AS path is announced to a prefix, we measure its difference in comparison with the previously stored path by calculating the edit distance. This metric, which is also called Levenshtein distance, is a sequence comparison method that indicates the minimum number of single-character edits (i.e. insertions and deletions) required in order to turn one sequence into the other. The average edit distance is calculated over all path changes.	AS-path
23	Maximum edit distance	This feature stores the highest edit distance in a given time window w .	AS-path
24-34	Edit distance with k value	Counts the number of message updates with a given $k(k = 0, \dots, 10)$ distance with respect to the previous known route.	AS-path
35-45	Edit distance with k value (unique)	Similar to the previous feature, but removes duplicate ASes in the path.	AS-path
46	Number of rare ASes	Keep track of how many times each AS appears in all the announced paths. Usually, an AS appears multiple times in a given time window. We classify an AS as rare if its number of appearances is below a given percentage threshold of ASes appearances (we set the default at the 95th percentile).	AS-path
47	Maximum number of rare ASes	The maximum number of ASes that appear in a message during a given time window w .	AS-path
48	Average number of rare ASes	The average number of ASes that appear in a given time window w .	AS-path

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